



DAVID K. TAN

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CLEARANCE

TS/SAP Fully Adjudicated | Active Secret Clearance

TECHNICAL SKILLS

- Modeling & Simulation: 6-DOF Flight Dynamics, Missile Simulation, and Monte Carlo Analysis.
- Programming: C++, Python, Java.
- DevOps & Tooling: GitLab CI/CD, Jenkins and Docker.
- Guidance & Control: Autopilot Implementation and Control Logic Integration.
- Software Engineering: Object-Oriented Design (OOD), Agile Development.
- Systems & Environment: Linux/Unix, Red Hat Enterprise Linux (RHEL).

PROFESSIONAL EXPERIENCE

6-DOF Modeling & Simulation Engineer, I3, Mar 2026 - Current

- Developed and validated high-fidelity 6-DOF missile simulation models (C++), including implementation of autopilot control logic for LRPM and integration of guidance, sensor, and flight dynamics components across programs including Hydra-70 and APKWS; increased simulation throughput by 30%, reducing validation cycle time and enabling faster performance assessment.
- Executed verification and validation (V&V) simulation studies, translating requirements into testable scenarios and running Monte Carlo analyses to evaluate system performance; identified model discrepancies and improved accuracy, reducing rework and increasing confidence in mission-critical results.
- Collaborated with cross-functional engineering teams to refine model assumptions, troubleshoot simulation issues, and align results with system requirements; improved model reliability and reduced debugging/iteration time, supporting more efficient analysis cycles.

Software Engineer, ALKU, Sep 2024 - Jan 2026, Hill AFB, UT

- Engineered high-fidelity 6-DOF missile flight dynamics simulations (C++, Python) in support of U.S. Air Force programs including the LGM-35A Sentinel program, applying numerical integration and sensor modeling to increase trajectory prediction accuracy by 25% and strengthen confidence in performance assessments.
- Diagnosed and resolved simulation inconsistencies by analyzing flight behavior and refining underlying model assumptions, improving model fidelity and reducing downstream validation effort.
- Streamlined integration and release processes by implementing GitLab CI/CD pipelines (YAML), enhancing build reliability and accelerating iteration cycles for simulation updates.

Software Engineer II, Space Dynamics Laboratory, Jan 2023 - Sep 2024, Logan, UT

- Implemented high-fidelity 6-DOF missile simulation components (C++, Python) within existing flight dynamics frameworks, contributing to U.S. Air Force programs including the FORGE program and supporting trajectory prediction improvements of up to 25% through physics-based modeling and numerical methods.
- Supported integration and verification of simulation models, analyzing outputs to identify inconsistencies and assist in refining system behavior, improving overall model stability and reliability.

Computer Scientist Intern, Bowie State University, Jun 2022 - Aug 2022, Remote

- Led cybersecurity research within a distributed systems REU program by designing and executing simulated network attacks, improving threat detection effectiveness by 28% through enhanced traffic analysis and anomaly modeling.

TECHNICAL LEADERSHIP

- Supported cross-functional integration between software, simulation, and mission operations teams in classified defense programs, improving coordination across model development, test planning, and mission analysis workflows.

EDUCATION

Master of Science, Computer Science	Jul 2027
Georgia Institute of Technology - Atlanta, GA	
MBA, Master of Business Administration	Dec 2023
Louisiana State University - Shreveport, LA	
Bachelor of Science, Computer Science	Dec 2022
Valdosta State University - Valdosta, GA	

CERTIFICATIONS

CompTIA Security+ | AWS Solutions Architect | CKA | Google Cloud Architect | CSEP

TECHNICAL LEADERSHIP

- Supported cross-functional integration between software, simulation, and mission operations teams in classified defense programs, improving coordination across model development, test planning, and mission analysis workflows.